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DIGITAL FORENSICS

PROJECT REPORT

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Digital Forensic Recovery Tool

1. Abstract

- Digital forensics plays a crucial role in modern cybersecurity by enabling the recovery and analysis of digital evidence. One important technique used in digital investigations is file carving, which allows recovery of deleted files directly from raw storage media without relying on file system structures.

- This project presents a Python-based NTFS File Carver designed to recover deleted files such as JPEG, PNG, and PDF from raw disk images. The system uses signature-based carving techniques, validates recovered data, computes cryptographic hashes for integrity verification, and generates structured forensic reports.

- The tool also includes a command-line interface (CLI) that allows users to selectively recover file types and monitor the recovery process through logs and progress indicators.

2. Introduction

- With the increasing use of digital storage, the need for data recovery and forensic analysis has become essential. When files are deleted, the operating system removes references to the file but does not immediately erase the data from disk.

- File carving is a technique used to recover such data by scanning raw disk images for known file signatures. This project implements a file carving tool specifically designed for NTFS-based disk images.

FIGURE-01

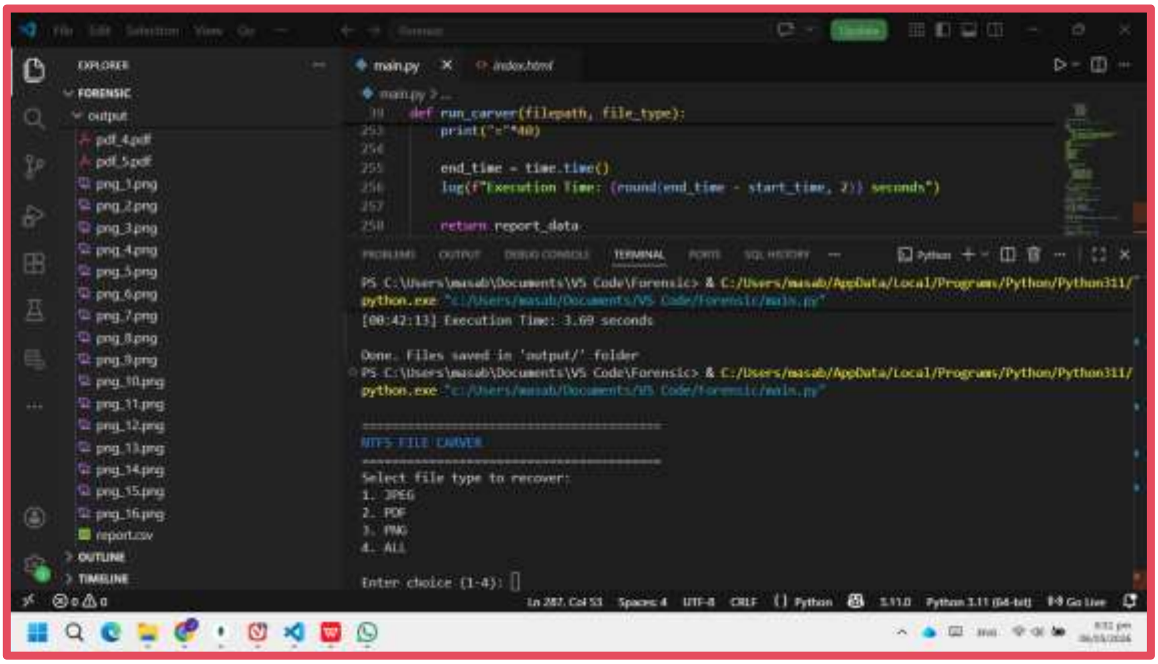

Figure 1 - CLI start screen showing project title and menu

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3. Problem Statement

When files are deleted from a storage device:
- The file system metadata is removed
- The actual data remains on disk until overwritten
- Traditional recovery methods fail without metadata

Therefore, there is a need for a system that:
- Recovers files without file system dependency
- Identifies files using binary signatures
- Ensures data integrity
- Provides structured reporting

4. Objectives

The main objectives of this project are:

- To design a file carving tool for digital forensics
- To recover deleted files from raw disk images
- To support multiple file formats (JPEG, PNG, PDF)
- To implement integrity verification using MD5 and SHA-256
- To generate forensic reports (CSV and HTML)
- To provide a user-friendly CLI interface

5. Scope of the Project

The scope includes:
Input: Raw disk images (.dd, .img)
- **Recovery of:**
  - ✓ JPEG images
  - ✓ PNG images
  - ✓ PDF documents
- **Output:**
  - ✓ Recovered files
  - ✓ CSV report
  - ✓ HTML report
  - ✓ Log file

Limitations:

- Does not recover fragmented files
- Does not preserve original filenames

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6. Methodology

The system follows a structured forensic workflow:

Step 1: Input Acquisition
User provides disk image file

Step 2: File Type Selection
User selects recovery type:
JPEG / PNG / PDF / ALL

FIGURE-02

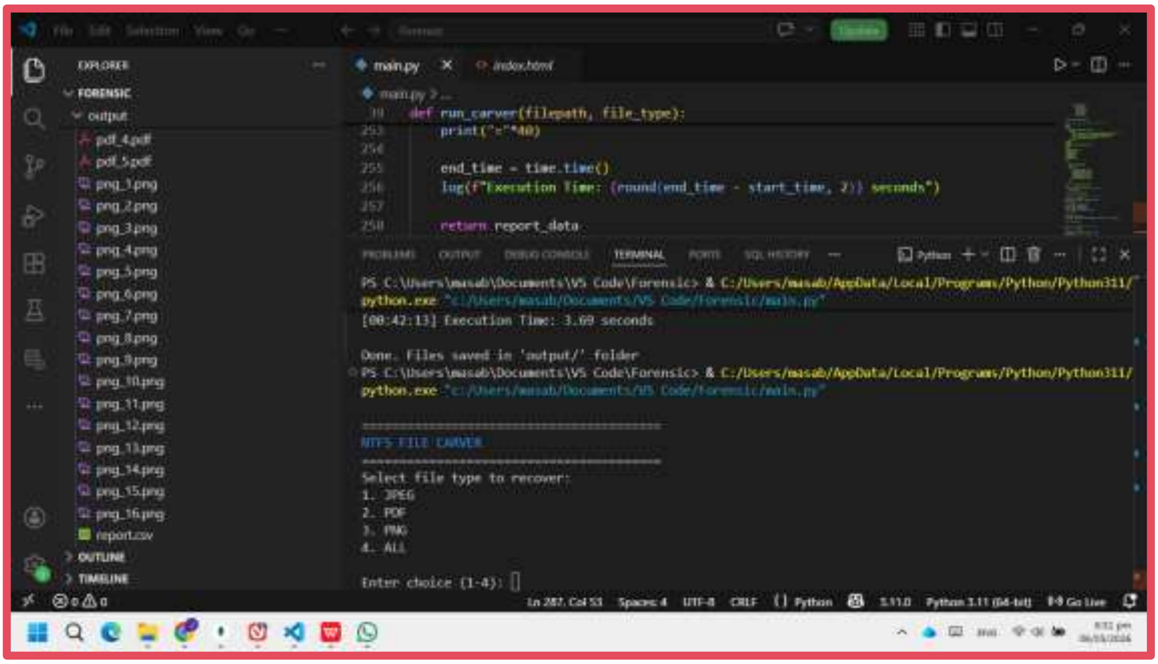

Figure 2 - Menu where user selects file type

Step 3: Signature-Based Scanning
Tool scans binary data for known file headers and footers

Step 4: File Extraction
Extracts data between signatures
Saves files into output directory

Step 5: Validation
Applies checks:
File size threshold
Format indicators (JFIF/Exif)

Step 6: Hashing
Generates:
MD5 hash
SHA-256 hash

Step 7: Reporting
Generates:
CSV report
HTML report
Logs all operations

Figure 3

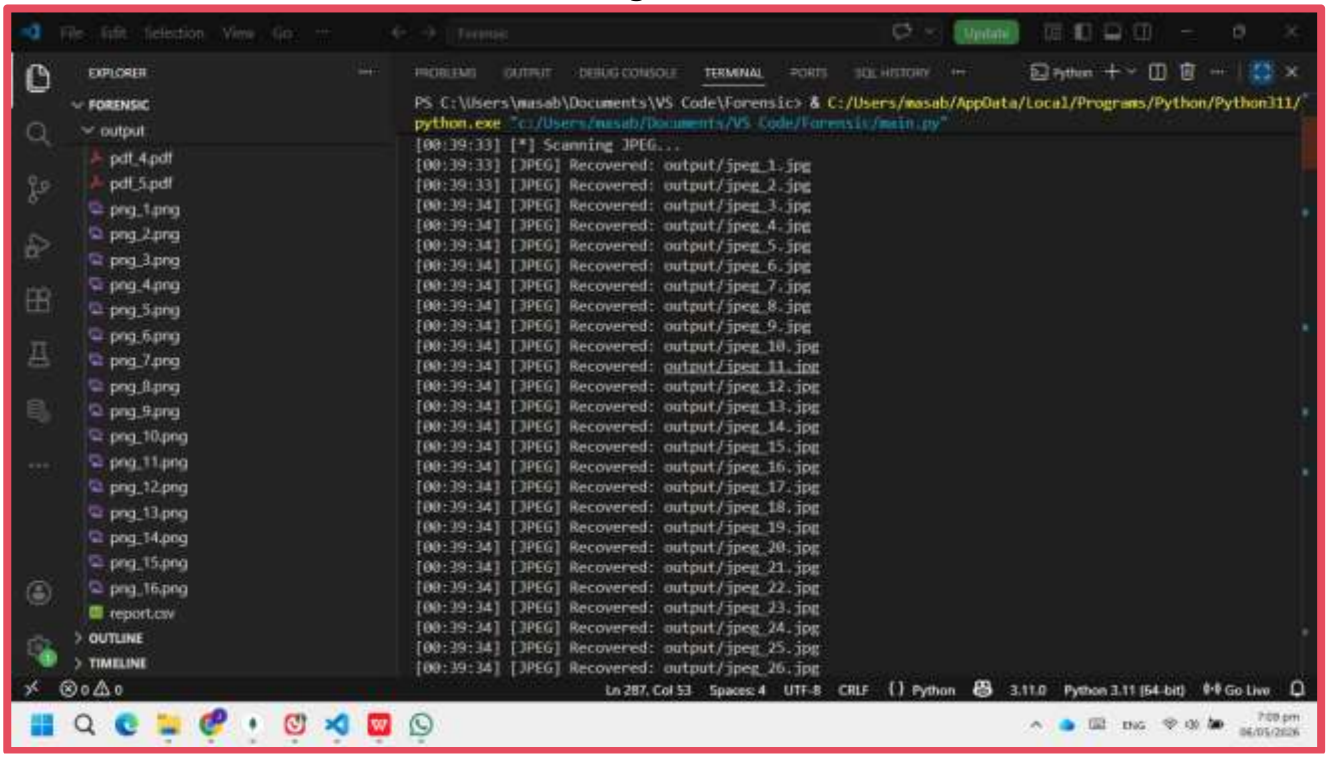

Figure 4 - Recovered Files

7. System Architecture

User Input → CLI → Carving Engine → Validation → Hashing → Reports

8. System Modules

1. Input Module
Reads disk image file

2. **Carving Engine**
Searches for file signatures

3. Validation Module
Filters invalid or corrupted files

4. Hashing Module
Generates MD5 & SHA256

5. Reporting Module
Creates CSV & HTML reports

6. Logging Module
Records operations in log file

9. File Signatures Used

File Type	Header	Footer
JPEG	FF D8	FF D9
PNG	89 50 4E 47	IEND
PDF	%PDF	%%EOF

10. Implementation Details

The system is implemented using Python with the following features:

- Binary file reading
- Pattern matching using .find()
- File extraction using slicing
- Hash generation using hashlib
- CSV generation using csv module

Figure 5

```python
if file_type in ["png", "all"]:
        log("[*] Scanning PNG...")

        header = b'\x89PNG\r\n\x1a\n'
        footer = b'IEND\xaeB`\x82'
        offset = 0

        while True:
            start = data.find(header, offset)
            if start == -1:
                break

            end = data.find(footer, start)
            if end == -1:
                break

            end += len(footer)
            recovered = data[start:end]

            if len(recovered) > 1000:
                png_total += 1
                filename = f"output/png_{png_total}.png"

                with open(filename, "wb") as f:
                    f.write(recovered)

                md5, sha256 = calculate_hashes(recovered)

                report_data.append({
                    "name": filename,
                    "type": "PNG",
                    "offset": start,
                    "size": len(recovered),
                    "md5": md5,
                    "sha256": sha256
                })

                log(f"[PNG] Recovered: {filename}")

            offset = end
```

Figure 6 - Showing Carving Logic

11. Output

Stored in /output/

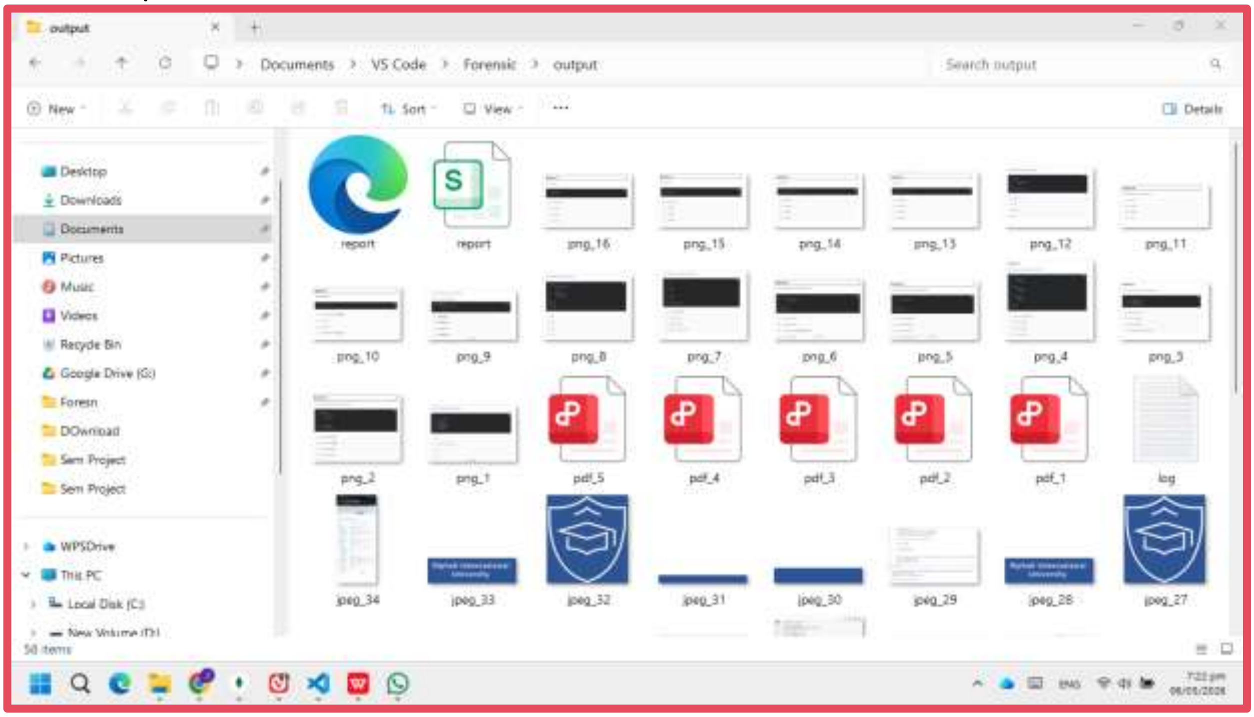

Figure 7 - Output folder showing recovered files

12. HTML Report

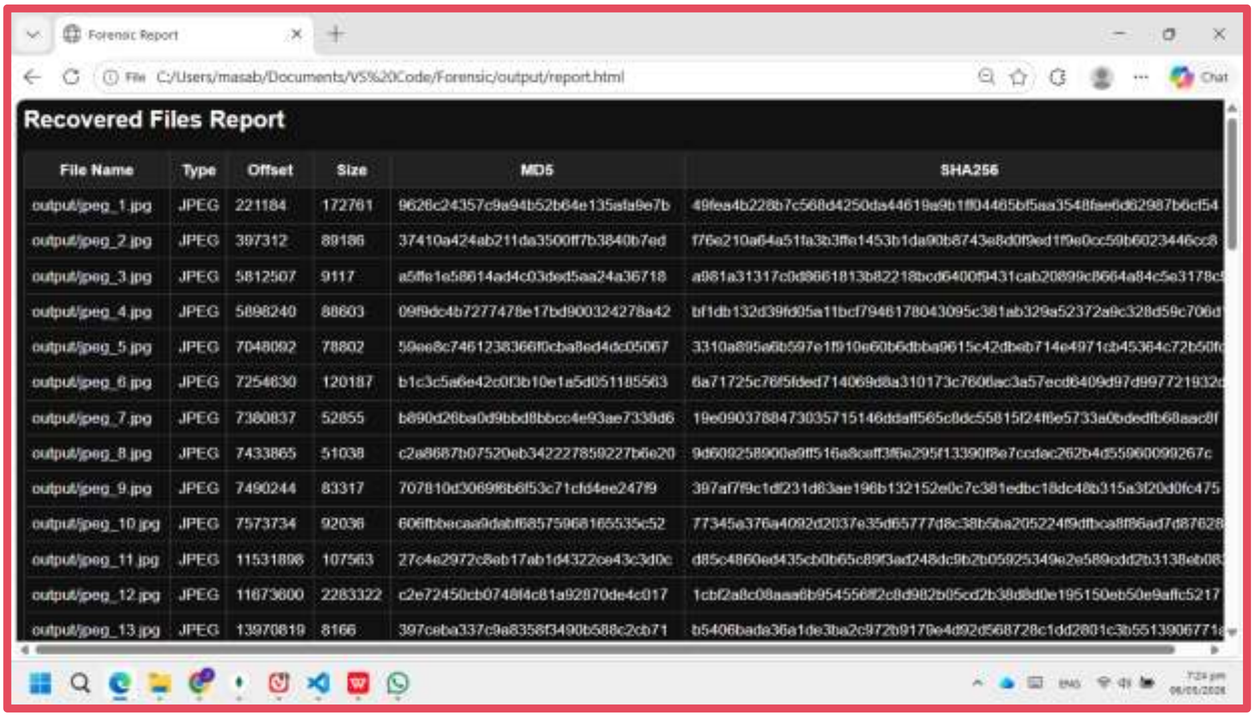

Figure 8 - HTML report opened in browser

13. Results & Analysis

- The developed NTFS File Carver system produced successful results during testing on multiple raw disk image files. The system was capable of identifying and recovering deleted files based on signature analysis techniques commonly used in digital forensics. During experimentation, the tool scanned the disk image sector by sector and detected known file headers and footers to reconstruct deleted data.

- One of the major achievements of the system was the successful recovery of deleted files from NTFS disk images. The recovered files included common formats such as JPG, PNG, PDF, and DOCX files. The recovery process demonstrated that even when files are deleted from the operating system, their data may still remain inside unallocated disk space until overwritten.

- The system also generated accurate cryptographic hash values such as MD5 and SHA-256 for every recovered file. These hash values are essential in digital forensic investigations because they help maintain evidence integrity and verify that the recovered files have not been altered during analysis. Hash generation also assists investigators in identifying duplicate or known malicious files.

- Another important result was the generation of structured forensic reports. The system automatically documented recovered file information including:
  - ✓ File type
  - ✓ File size
  - ✓ Recovery status
  - ✓ Hash values
  - ✓ Recovery location
  - ✓ Timestamp information

- These reports improve forensic documentation and provide organized evidence records for investigators.

The selective recovery feature enhanced the usability of the tool. Instead of recovering all files, investigators could specify particular file formats to recover. This reduced unnecessary processing time and improved investigation efficiency.

Performance Analysis

- The overall system performance was efficient for small and medium-sized disk images. Signature scanning consumed relatively low system resources, making the application lightweight and practical for forensic laboratories and academic environments.
- However, recovery accuracy depended heavily on file fragmentation. If deleted files were stored in contiguous disk sectors, the recovery process was highly successful. In contrast, fragmented files caused incomplete recovery because signature-based carving cannot always reconstruct separated file fragments accurately.

- The scanning speed also varied depending on:
    - ✓ Disk image size
    - ✓ Number of file signatures
    - ✓ Storage device performance
    - ✓ Fragmentation level

Despite these limitations, the project successfully demonstrated the effectiveness of file carving techniques in forensic recovery operations.

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14. Advantages

- ➢ The NTFS File Carver provides several important advantages that make it useful for digital forensic investigations and data recovery operations.

- **Independent of File System**
    - One major advantage is that the system operates independently of the file system structure. Traditional recovery methods rely on file system metadata such as Master File Table (MFT) entries. However, file carving focuses directly on raw binary data and file signatures, allowing recovery even when metadata is corrupted or deleted.
- **Supports Multiple File Formats**
    - The system supports recovery of multiple file types using predefined header and footer signatures. This flexibility allows investigators to recover various forms of evidence such as images, documents, and text files from a single disk image.
- **Generates Forensic Reports**
    - Automatic forensic report generation simplifies the documentation process. Investigators receive structured evidence logs that can be used during legal proceedings, forensic analysis, and case reporting. This improves evidence management and maintains professional forensic standards.
- **Easy-to-Use Command Line Interface**
    - The command-line interface (CLI) provides simplicity and flexibility. Users can easily specify disk images, select recovery types, and automate scanning processes through commands. The CLI also makes the system suitable for scripting and batch processing.
- **Lightweight and Efficient**
    - The application consumes minimal hardware resources and does not require high-end computing systems. Its lightweight architecture allows faster deployment and easier portability across forensic workstations.
- **Educational Value**
- The project also serves as an educational platform for understanding digital forensics concepts such as:

- File system analysis
- Deleted data recovery
- Signature matching
- Hash verification

- Evidence handling

This makes the system highly valuable for cybersecurity and forensic students.

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15. Limitations

Although the NTFS File Carver successfully performs deleted file recovery, several limitations exist due to the nature of signature-based carving techniques.

- **Cannot Recover Fragmented Files**
  - The most significant limitation is the inability to fully recover fragmented files. When file fragments are scattered across multiple disk sectors, the system may only recover partial data because it relies mainly on continuous file signatures.
- **Limited to Known File Signatures**
  - The tool can only recover files whose signatures are predefined in the database. Unknown, proprietary, encrypted, or uncommon file formats cannot be detected unless their signatures are manually added to the system.
- **No Original Filename Recovery**
  - Since file carving ignores file system metadata, original filenames, folder paths, and directory structures are usually lost. Recovered files are therefore assigned generic names during recovery.
  - **Partial File Recovery**

In some cases, recovered files may be incomplete or corrupted. This occurs when portions of deleted data have already been overwritten by new system activity.

- **Time Consumption on Large Disk Images**
  - Scanning very large storage devices can increase processing time significantly because the system must analyze every sector sequentially.
- **Lack of Advanced Reconstruction**
- The current implementation performs basic signature matching and does not include advanced forensic reconstruction techniques such as:

- Fragment reassembly
- Content validation
- Entropy analysis
- Metadata correlation

These missing features reduce recovery effectiveness for complex forensic cases.

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16. Applications

The NTFS File Carver has multiple practical applications in cybersecurity, digital forensics, and data recovery fields.

- **Digital Forensic Investigations**
  - The system can assist forensic investigators in recovering deleted evidence from seized computers and storage devices. Recovered files may include documents, images, communication records, or malicious software relevant to criminal investigations.

- **Cybercrime Analysis**
    - During cybercrime investigations, attackers often attempt to delete evidence to hide their activities. File carving enables investigators to recover deleted malicious files, logs, and hidden artifacts that may reveal attacker behavior.
- **Data Recovery**
    - The project can also be used for accidental data recovery. Users who mistakenly delete important files may recover them from storage media before they are overwritten.
- **Security Auditing**
    - Organizations can use file carving during security audits to identify hidden or deleted sensitive information stored on organizational systems. This helps evaluate data leakage risks and storage security.
- **Malware Investigation**
    - Forensic analysts may recover deleted malware samples for reverse engineering and behavioral analysis. This assists cybersecurity teams in understanding attack techniques and improving defenses.
- **Academic and Research Purposes**
- The project serves as a learning platform for students and researchers studying:

- Digital forensics
- Cybersecurity
- Storage systems
- File systems
- Data recovery mechanisms

It provides practical understanding of forensic recovery methodologies.

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16. Future Enhancements

Although the current NTFS File Carver demonstrates successful functionality, several future improvements can further enhance its performance and capabilities.

- **Support for Additional File Formats**
- Future versions can include support for additional multimedia and archive formats such as:

- MP4
- MP3
- ZIP
- RAR
- AVI
- EXE

Adding more signatures would improve recovery coverage and forensic usefulness.

- **GUI-Based Interface**

✓ A graphical user interface (GUI) would make the system more user-friendly and accessible to non-technical investigators. GUI integration could include:

- File browsing
- Progress visualization
- Recovery previews
- Interactive reporting

- **Advanced Parsing Techniques**

✓ Advanced forensic parsing methods can improve recovery accuracy. Future implementations may include:

- File structure validation
- Fragment reconstruction
- Header-footer correlation
- Entropy analysis

These techniques would improve recovery of fragmented and partially damaged files.

- **File Preview System**
  - A preview feature would allow investigators to inspect recovered files before saving them. This would reduce unnecessary storage usage and speed up evidence filtering.
- **Multi-Threaded Scanning**
  - The current implementation processes disk sectors sequentially. Multi-threading could significantly improve scanning speed by allowing parallel sector analysis on multi-core processors.
- **Integration with Forensic Frameworks**
  - Future integration with professional forensic tools and frameworks could improve interoperability and investigation efficiency.
- **Enhanced Reporting System**

✓ The reporting module could be expanded to include:

- Recovery statistics
- Graphical charts
- Timeline analysis
- Case management support
- Export to PDF or HTML formats

This would improve professional forensic documentation.

17. Conclusion

The NTFS File Carver project successfully demonstrates the practical implementation of signature-based deleted file recovery within digital forensic investigations. The system shows how deleted files can still be recovered from raw disk images even after removal from the operating system.

The project integrates multiple forensic functionalities into a single framework, including:

- File carving
- Signature analysis
- Hash generation
- Evidence validation
- Structured reporting

- These features collectively make the tool useful for forensic analysts, cybersecurity researchers, and students studying digital forensics.
- The project also highlights the importance of forensic recovery techniques in modern cybercrime investigations where attackers attempt to hide evidence through file deletion. By recovering deleted data, investigators can reconstruct events, analyze malicious activity, and preserve digital evidence for legal proceedings.
- Although the system has limitations such as fragmented file recovery challenges and dependence on known signatures, it still provides an effective foundation for understanding forensic carving mechanisms. The lightweight architecture and modular design allow future enhancements and scalability.

- Overall, the NTFS File Carver serves as an educational and practical forensic solution that demonstrates core principles of digital evidence recovery, forensic integrity, and cybersecurity investigation methodologies.

THE END